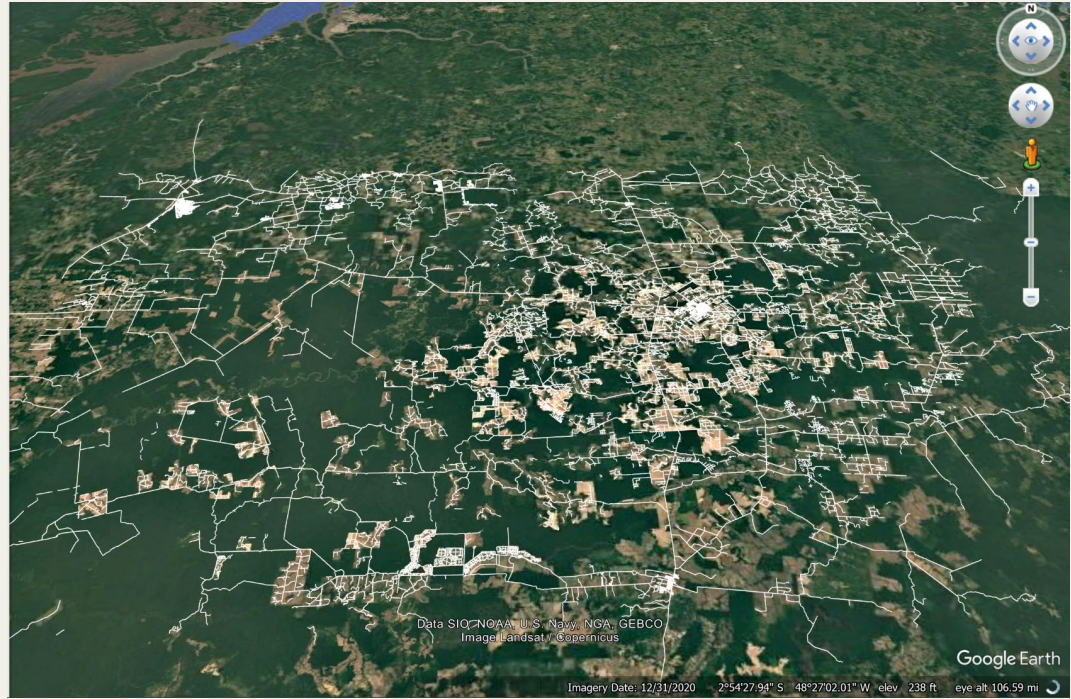


Mapping Paragominas with Discrete Graph Theory

By Caitlyn Feldpausch



Starting Point

■ Mapping the original datasets

- **OSM Dataset:**

Verified road networks

Contains detailed highway data.

- **PrevisIA Dataset:**

35,225 records, 35,791 lines and about 13,840 km of unconnected roads.

Derived from Sentinel-2 Imagery

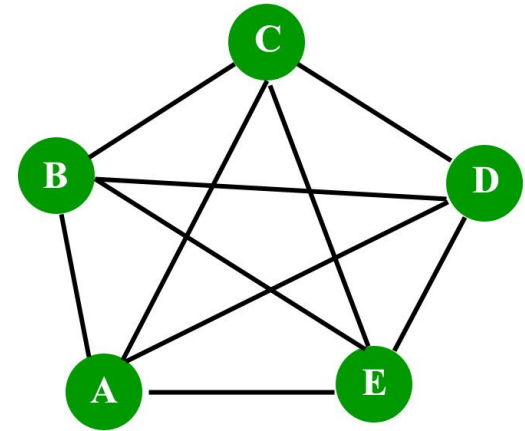
Both networks in CRS: WGS84



Goal:

To apply Discrete Graph Theory to combined and terrain-corrected versions of our datasets and discover what section of the road is an intersecting set, meaning the entire path is fully transversable.

Then export submapped, corrected data into a field map app for workers.



Complete Graph

Task One

1 Inspect Original Data in Google Earth Engine

Get an overall picture and understanding of the unconnected data before do any further analysis.

4 Map Paragominas Combined Dataset Highway Networks

Create a data visualization of the map broken down by types of highways or roads present.

This turned out to be a very messy map with many classifications, so...

2 Load and Merge the Datasets

Merge and geometrically snap both datasets into a single, unified, routable network that bridges official roads to remote, unmapped targets.

5 Reclassify Roadway Data

Create a cleaner visualization by reclassifying the detailed highway data column into AI or unclassified traces, paved or main roads, and rural or dirt OSM roads.

2 Coordinate Reference Systems and Data Inspection

Ensure both datasets are in WGS84.

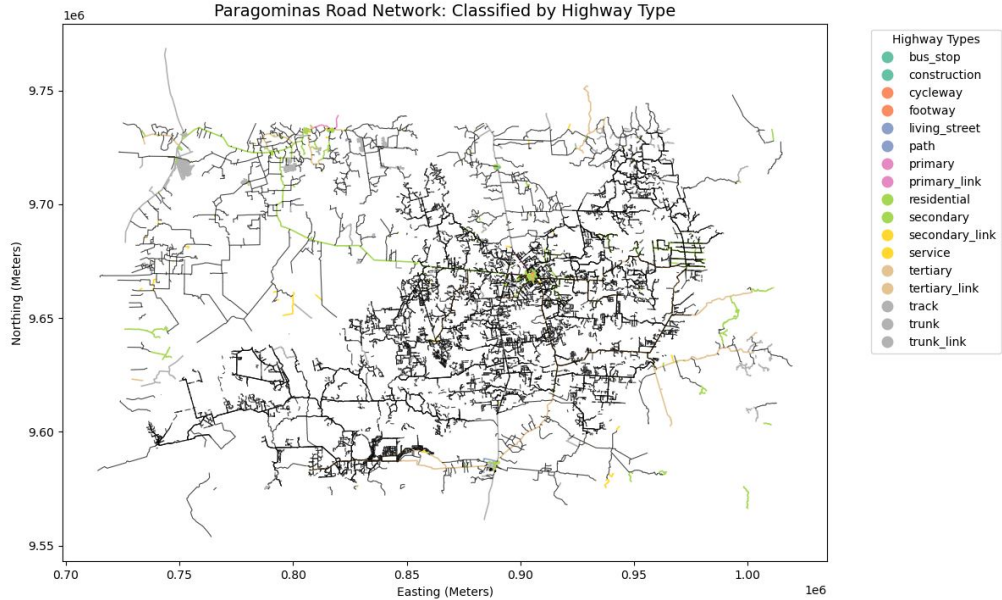
Inspect OSM highway and surface columns to understand OSM network features.

Handle missing values and messy categorical data.

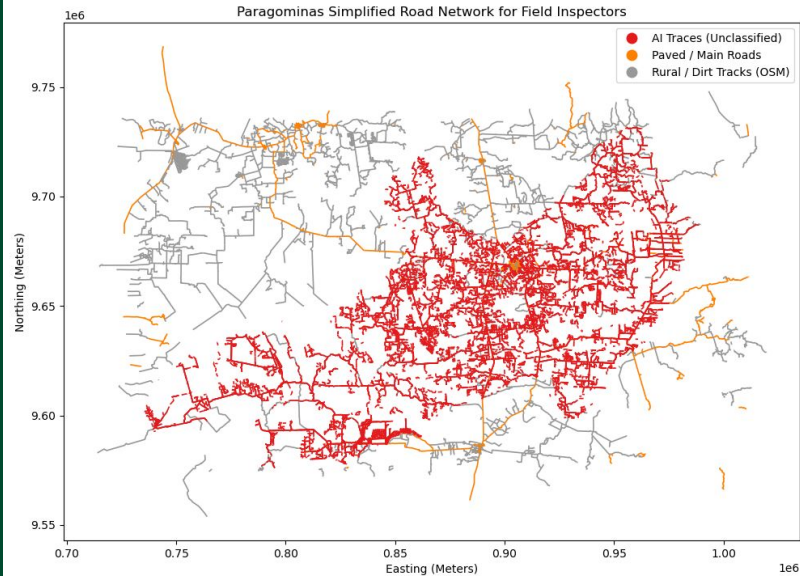
6 Export Data for Next Task

Export the more detailed, combined dataset with many highway classes to increase analysis accuracy in next tasks.

Paragominas Road Network: Classified by Highway Type



Paragominas Simplified Road Network for Field Inspectors



Task Two

1 Cost Functions to Weight The Discrete Graphs

Calculate Weights or Cost Functions for the discrete graph nodes, or the points where roads connect.

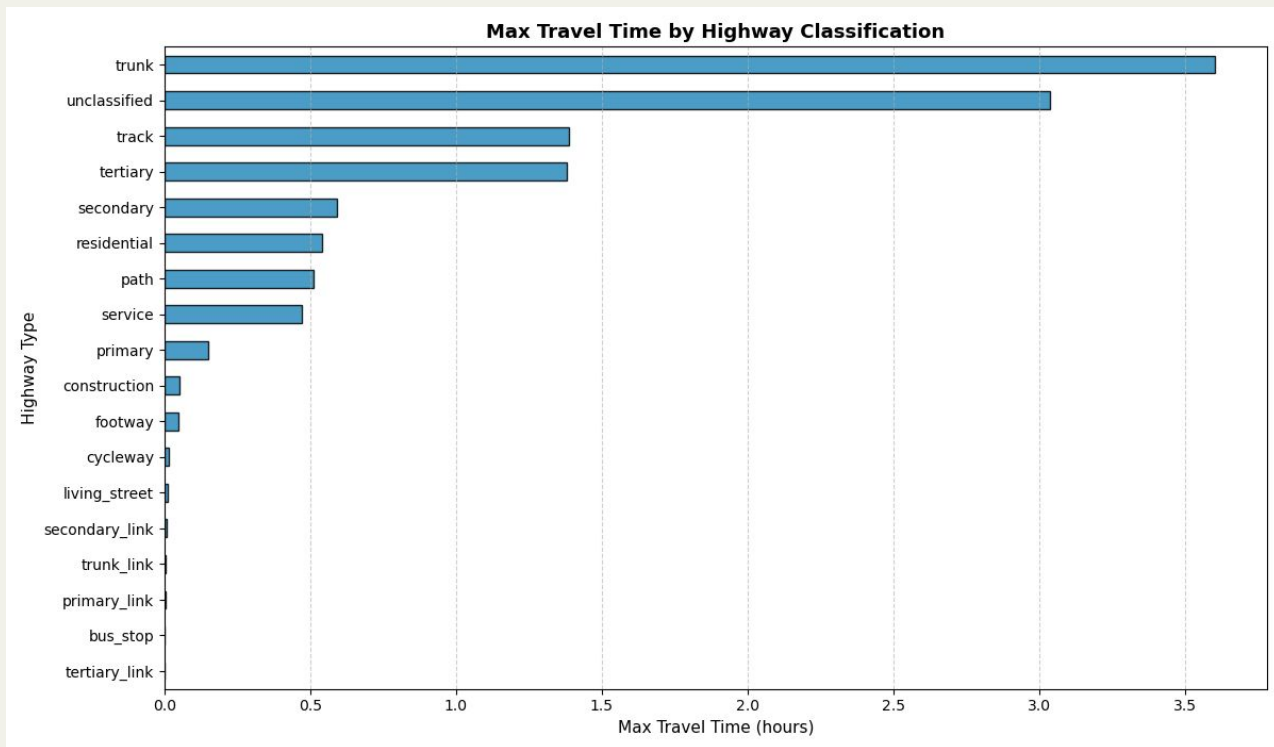
Do this by assigning an estimated speed in km/h based on each segment's highway tag via a custom dictionary lookup,

Then calculate the length of every road segment in meters.

Derive the total travel time in hours with $\text{Time} = \text{Distance} / \text{Speed}$.

2 Map Travel Times by Highway Type

3 Export the dataset with computed travel times for the next task.



Task Three

1

Create a Spatial Index and Snap Topology Together

Separate the OSM and PrevisIA data.

Build a spatial index (or R-tree) on the OSM dataset for high-quality, verified lookups.

Iterate through each PrevisIA trace to "snap" its geometric nodes or endpoints to nearby official OSM roads using a boundary or tolerance of 15-meters and shapely.

2

Recombine the "Snapped Together" Dataset and Export

Recombine the newly snapped PrevisIA traces with the original OSM data into a unified Geospatial Data Frame

Save one version for the upcoming NetworkX graph step.

Reproject another version back to WGS84 Lat/Lon data and export it as a GeoJSON for future app work.

Task Four

1

Data Cleaning Again

Parse raw GeoJSON datasets where separate lines are stored independently and their coordinates are misaligned even if they appear to connect fully on the screen.

Topographic correction fixes this as an additional step to mathematically lock nodes or road end points together.

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Smaller roads sprout from Amazonian highways in a deforestation process known as a fishbone pattern. Image courtesy of NASA Landsat.

Task Five/Six: Graph Modeling & NetworkX Integration: Dijkstra's algorithm

1

Convert snapped spatial geometries into a NetworkX, multidimensional graph structure

Use ordered coordinate pairs (x, y) as node identifiers so multiple converging road segments can intersect and share the exact same point.

Weight edges or the lines connecting nodes by travel time and include lengths and highway classifications:

Node edges (lines connecting them) will be weighted by travel times.

Edge lengths will be the highway length in meters.

Lines between nodes will be highways.

Discard isolated forest stubs, dead ends, and disjointed components, making a submap of fully connected roads.

2

Field Routing & KD-Tree Spatial Optimization

Create a reusable field-mapping framework using a SciPy KDTree spatial index coupled with pyproj coordinate transformers.

This handles fragile networks that can change fast from broken roads.

Faster spatial lookups snap GPS input coordinates to the nearest clean graph nodes.

A function then calculates the best route path, extracts turns, and exports clean, field-ready GeoJSON routes for workers. .

Task Seven

1

Use Google Earth Engine and QGIS to Map Creates GeoJSON Exports

Script in Google Earth Engine Code Editor to map high-resolution Sentinel-2 Satellite Imagery Surface Reflectance Level data for our app basemap aligned with where the PrevisIA data came from.

A cloud mask filtered cloud cover to 20%.

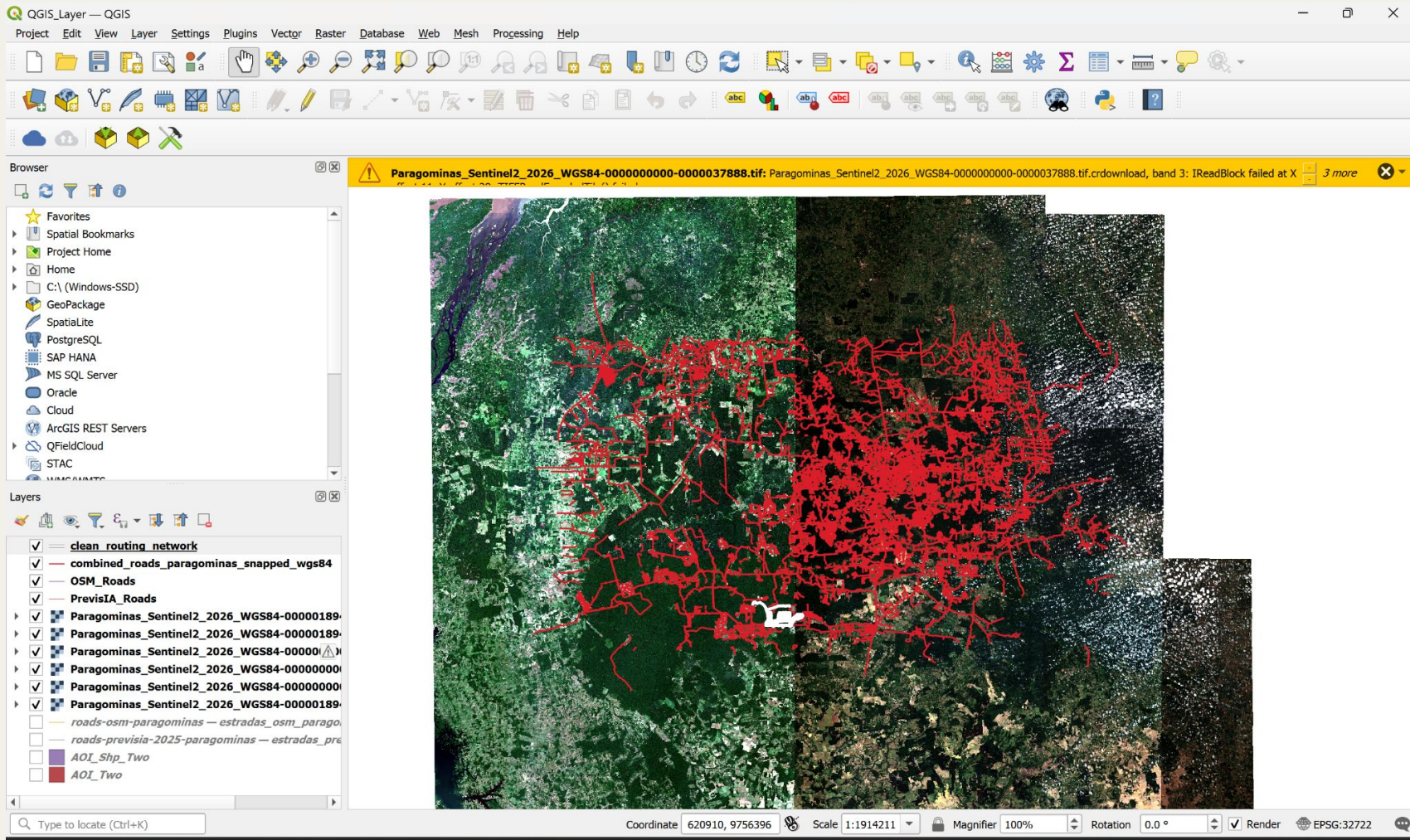
Used summer months of 2026 to get clear sky data.

Exporter raster layer to drive before importing into QGIS.

2

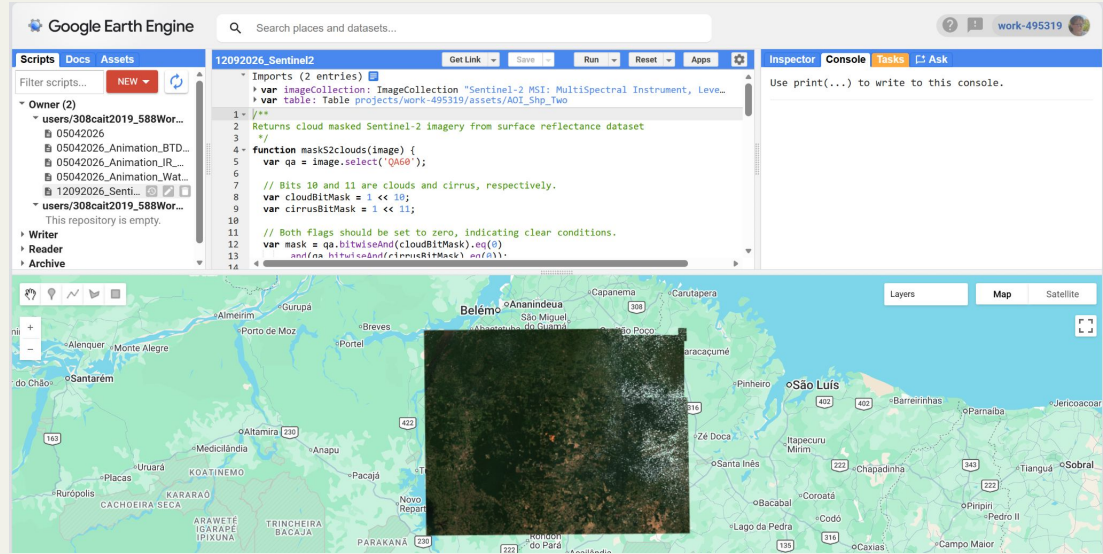
Export QGIS Package Into QField for Offline Analysis

Export QGIS Layers (old data, new cleaner and combined data, the submap of fully connected roads, and a sample route on that submap with the Sentinel-2 imagery as a baselayer.



Conclusions:

- 95% of roads mapped are disjoint sets in graph theory, or unconnected.
- This corroborates what we know about the disjointed data provided and that many Amazon roads are disconnected for nefarious purposes like mining, deforestation.



<https://code.earthengine.google.com/1402379357a5e2e9b59210ad59abc391>

Conclusions:

- Mapping our final layers on Android was easy.
- Mapping them on my iPad changed my symbology and stopped me from mapping both the raster Sentinel-2 high-resolution layer with my road networks.



Further Analysis:

- Utilize the QField Package to edit these cleaned layers as more roads become connected.
- Utilize field worker insights and drone data collections methods to continue optimizing the field app.



Link to Related Article:

<https://news.mongabay.com/2026/07/crackdown-lets-rainforest-reclaim-illegal-road-in-rare-win-for-the-amazon/>

Satellite imagery confirmed that a 42.8-kilometer (26.6-mile) illegal road in Pará has completely disappeared,